

$$\frac{F_k}{r_+^k} \longrightarrow c_+ \quad (\text{homework})$$

$$F_k \text{ grows like } c_+ r_+^k = c_+ \left(\frac{1 + \sqrt{5}}{2} \right)^k.$$

Theorem. Let

$$\text{Fib} = \{(x_k)_{k \geq 0} : x_{k+2} = x_{k+1} + x_k\}.$$

1. Fib is a subspace of S .
2. The map $T : \text{Fib} \rightarrow \mathbb{R}^2$, given by

$$T((x_k)) = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix},$$

is linear and invertible.

3. $\{(r_+^k), (r_-^k)\}$ is a basis in Fib, where

$$r_{\pm} = \frac{1 \pm \sqrt{5}}{2}.$$

4. Any vector (x_k) in Fib can be written as

$$c_+(r_+^k) + c_-(r_-^k),$$

where

$$\begin{bmatrix} c_+ \\ c_- \end{bmatrix}$$

are coordinates of (x_k) in the basis $\{(r_+^k), (r_-^k)\}$.

Generalization to 2nd-order recursions

Let (x_k) be a sequence in S such that

$$x_{k+2} + a_1 x_{k+1} + a_0 x_k = 0 \quad \text{for all } k, \quad a_0 \neq 0.$$

Theorem. Such sequences form a 2-dimensional subspace H of S , the vector space of all sequences.

Proof. Let $T : H \rightarrow \mathbb{R}^2$ be

$$T((x_k)_{k \geq 0}) = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}.$$

Then T is linear and invertible.

How to find solutions. Consider

$$x_{k+2} + a_1 x_{k+1} + a_0 x_k = 0, \quad a_0 \neq 0.$$

Search for solutions of the form $(r^k)_{k \geq 0}$ for some $r \neq 0$. Then

$$r^{k+2} + a_1 r^{k+1} + a_0 r^k = 0.$$

Dividing by r^k gives

$$r^2 + a_1 r + a_0 = 0,$$

so

$$r = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_0}}{2}.$$

Case 1: $a_1^2 > 4a_0$.

The two nonzero roots r_+ and r_- are two independent vectors in the 2-dimensional space H . They form a basis, so any vector in H is

$$c_+(r_+^k) + c_-(r_-^k).$$

Example. $a_1 = -1 = a_0$: Fibonacci.

Case 2: $a_1^2 = 4a_0$.

$$r = -\frac{a_1}{2} \neq 0.$$

Then $(r^k)_{k \geq 0} = \left((- \frac{a_1}{2})^k\right)_{k \geq 0}$ solves the recursion. In this case, another vector is needed to form a basis in H .

Try

$$(x_k)_{k \geq 0} = (kr^k)_{k \geq 0}, \quad r = -\frac{a_1}{2}.$$

Substitution.

$$\begin{aligned} & (k+2)r^{k+2} + a_1(k+1)r^{k+1} + a_0kr^k \\ &= r^k [(k+2)r^2 + a_1(k+1)r + a_0k] \\ &= r^k \left[(k+2)\frac{a_1^2}{4} - \frac{a_1^2}{2}(k+1) + \frac{a_1^2}{4}k \right] \\ &= r^k \frac{a_1^2}{4} [k+2 - 2(k+1) + k] = 0. \end{aligned}$$

Here $r = -a_1/2$ and $a_0 = a_1^2/4$.

Theorem. If $r^2 + a_1r + a_0 = 0$, $a_0 \neq 0$, has a repeated root

$$r = -\frac{a_1}{2} \neq 0,$$

then $\{(r^k), (kr^k)\}$ is a basis in the solution space H .

Example.

$$x_{k+2} + 2x_{k+1} + x_k = 0, \quad r^2 + 2r + 1 = 0, \quad r = -1.$$

Thus

$$\{(-1)^k, k(-1)^k\}$$

is a basis in H .

Case 3: $a_1^2 < 4a_0$.

The equation $r^2 + a_1r + a_0 = 0$ has two complex roots:

$$r_{\pm} = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_0}}{2} = \frac{-a_1 \pm i\sqrt{4a_0 - a_1^2}}{2}.$$

Then (r_+^k) and (r_-^k) solve

$$x_{k+2} + a_1x_{k+1} + a_0x_k = 0.$$

Example. If $a_1 = 0$ and $a_0 = 1$, then

$$r_{\pm} = \pm i,$$

and $(i^k), ((-i)^k)$ solve the recursion

$$x_{k+2} + x_k = 0.$$

Also,

$$i^k = (i, -1, -i, 1, i, i^2, i^3, i^4, \dots).$$

If r_+^k, r_-^k are nonreal solutions of

$$x_{k+2} + a_1x_{k+1} + a_0x_k = 0,$$

then

$$\frac{r_+^k + r_-^k}{2}, \quad \frac{r_+^k - r_-^k}{2i}$$

are real solutions.

Proof. Since $r_- = \overline{r_+}$,

$$r_-^k = \overline{r_+^k}.$$

Writing $r_+^k = a + ib$ gives

$$\frac{r_+^k + r_-^k}{2} = \frac{a + ib + a - ib}{2} = a,$$

and

$$\frac{r_+^k - r_-^k}{2i} = \frac{a + ib - (a - ib)}{2i} = b.$$

Theorem. The two real solutions

$$\left(\frac{r_+^k + r_-^k}{2} \right), \quad \left(\frac{r_+^k - r_-^k}{2i} \right)$$

are independent in S .

Proof. Let $u = (r_+^k)$. Then

$$(r_-^k) = \overline{(r_+^k)} = \overline{u}.$$

Suppose there is a real number c with

$$\frac{u + \overline{u}}{2} = c \frac{u - \overline{u}}{2i}.$$

Then

$$iu + i\overline{u} = cu - c\overline{u},$$

so

$$(c + i)\overline{u} = (c - i)u = (c - i)(r_+^k).$$

Thus

$$\left(\frac{r_+}{r_-} \right)^k = \frac{c + i}{c - i}$$

is constant, which is impossible.

Theorem. If r_+, r_- are nonreal distinct roots of

$$r^2 + a_1 r + a_0 = 0,$$

then

$$\left\{ \frac{r_+^k + r_-^k}{2}, \frac{r_+^k - r_-^k}{2i} \right\}$$

is a basis in the solution space of

$$x_{k+2} + a_1 x_{k+1} + a_0 x_k = 0.$$

Generalization

Let A be an $n \times n$ real matrix. Let $(X_k)_{k \geq 0}$ be a sequence of vectors in $\mathbb{R}^n$.

Let S^n be the vector space of sequences of vectors in $\mathbb{R}^n$:

$$(X_1, AX_1, A^2 X_1, A^3 X_1, \dots).$$

Recursion.

$$X_{k+1} = AX_k.$$

Example. Write

$$x_{k+2} + a_1 x_{k+1} + a_0 x_k = 0$$

in matrix form. Let

$$A = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix}, \quad X_k = \begin{bmatrix} x_k \\ x_{k+1} \end{bmatrix}, \quad X_{k+1} = \begin{bmatrix} x_{k+1} \\ x_{k+2} \end{bmatrix}.$$

Then

$$X_{k+1} = AX_k = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} \begin{bmatrix} x_k \\ x_{k+1} \end{bmatrix} = \begin{bmatrix} x_{k+1} \\ -a_0 x_k - a_1 x_{k+1} \end{bmatrix}.$$

Theorem. Let A be an $n \times n$ real matrix. The solution space of

$$X_{k+1} = AX_k$$

has dimension n .

Proof.

$$X_{k+1} = AX_k = AAX_{k-1} = \dots = A^k X_1.$$

Let $T : S^n \rightarrow \mathbb{R}^n$ be the map sending the sequence $(X_k)_{k \geq 1}$ to X_1 . The map is invertible, since the linear map T is a linear isomorphism.

Notation.

$$z = a + ib, \quad a, b \text{ real.}$$

One can do linear algebra with matrices whose entries are complex numbers:

$$\mathbb{C}^n = \{\text{all } n \times 1 \text{ matrices with complex entries}\}.$$

One can talk about vector spaces with complex scalars.

Examples. Polynomials with complex coefficients; sequences (x_k) where each x_k is a complex number.

Theorem. Complex conjugation commutes with addition and multiplication of complex numbers.

Corollary. Complex conjugation commutes with addition and multiplication of matrices.

Example.

$$\overline{\begin{bmatrix} i & 2+i \\ -1 & -1-i \end{bmatrix}} = \begin{bmatrix} -i & 2-i \\ -1 & -1+i \end{bmatrix}.$$

Proof. The ij entry of $\overline{A+B}$ is the conjugate of the ij entry of $A+B$, so

$$\overline{ij \text{ entry of } A + ij \text{ entry of } B} = ij \text{ entry of } \overline{A} + ij \text{ entry of } \overline{B}.$$

For multiplication, the ij entry of $\overline{AB}$ is

$$\begin{bmatrix} \overline{a_{i1}} & \cdots & \overline{a_{in}} \end{bmatrix} \begin{bmatrix} \overline{b_{1j}} \\ \vdots \\ \overline{b_{nj}} \end{bmatrix} = \overline{a_{i1}} \overline{b_{1j}} + \cdots + \overline{a_{in}} \overline{b_{nj}} = \overline{a_{i1}b_{1j} + \cdots + a_{in}b_{nj}},$$

which is the ij entry of $\overline{AB}$.

Example: n th-order linear recursion

Let (x_k) be a sequence of real numbers that solves

$$x_{k+n} + a_{n-1}x_{k+n-1} + \cdots + a_0x_k = 0, \quad a_0 \neq 0. \quad (*)$$

The space of such sequences has dimension n .

Let

$$X_k = \begin{bmatrix} x_k \\ x_{k+1} \\ \vdots \\ x_{k+n-1} \end{bmatrix}, \quad X_{k+1} = \begin{bmatrix} x_{k+1} \\ x_{k+2} \\ \vdots \\ x_{k+n} \end{bmatrix},$$

and

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-2} & -a_{n-1} \end{bmatrix}.$$

Then $X_{k+1} = AX_k$.

Theorem. The map $(X_k) \mapsto (x_k)$ is a linear isomorphism, so the solution space of $(*)$ has dimension n . We assume $a_0 \neq 0$ because ...

Example. What is the dimension of the solution space of

$$x_{k+4} - x_{k+2} + x_{k+1} = 0?$$

The index k appears in the recursion only through $x_{k+1}, x_{k+2}, x_{k+4}$, so x_0 never enters it. Shifting the index (working with $(x_k)_{k \geq 1}$) gives

$$x_{k+3} - x_{k+1} + x_k = 0,$$

a 3rd-order recursion with $a_0 = 1 \neq 0$, so the theorem applies:

$$\dim = 3.$$

Note. We need to learn more linear algebra to solve $X_{k+1} = AX_k$. We need to compute A^k for all k at once.